

Degree: B.Tech Semester: 3<sup>rd</sup> Sem

## END-SEMESTER EXAMINATION (SUMMER), 2026

Course Code: EAEPC302

Course Title: Signal and Systems

Duration: 03 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suit assumed & mentioned in the answer.

| Q. No. | Question                                                                                                                                                                          | Marks |  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--|
| Q 1    | Attempt any 2 parts of the following                                                                                                                                              |       |  |
| 1a     | Classify the following signals as:<br>Continuous-time or Discrete-time, Energy or Power, Even or Odd.<br>i. $x(t) = e^{-2t}u(t)$<br>ii. $x[n] = \sin\left(\frac{\pi n}{4}\right)$ | 5     |  |
| 1b     | Explain the transformation of independent variables (time shifting, time scaling and time reversal) with suitable examples.                                                       | 5     |  |
| 1c     | Define orthogonality of signals. Show that the functions $\sin n\omega_0 t$ and $\cos n\omega_0 t$ are orthogonal over one fundamental period.                                    | 5     |  |
| Q 2    | Attempt any 2 parts of the following                                                                                                                                              |       |  |
| 2a     | State and explain the properties of convolution. Discuss the significance of impulse response and step response of an LTI system.                                                 | 5     |  |
| 2b     | Explain the graphical method of convolution. Determine the output of an LTI system for<br>$x(t) = u(t) - u(t - 2), h(t) = e^{-t}u(t)$<br>using convolution integral.              | 5     |  |
| 2c     | Compute the convolution sum of $x[n] = \{2, 1, 3\}, h[n] = \{1, -1, 2\}$ .                                                                                                        | 5     |  |
| Q 3    | Attempt any 2 parts of the following                                                                                                                                              |       |  |
| 3a     | Derive the exponential Fourier Series representation of a continuous-time periodic signal. State any four properties of Fourier Series.                                           | 5     |  |
| 3b     | Define the Continuous-Time Fourier Transform (CTFT). State its important properties. Explain the magnitude and phase spectrum of a signal with a suitable example.                | 5     |  |
| 3c     | Differentiate between Low Pass, High Pass, Band Pass and All Pass filters. Explain the relationship between discrete-time signals and sampled continuous-time low-pass signals.   | 5     |  |
| Q 4    | Attempt any 2 parts of the following                                                                                                                                              |       |  |
| 4a     | Define the Laplace Transform. Explain the concept of Region of Convergence (ROC) and discuss its properties.                                                                      | 5     |  |
| 4b     | Determine the Laplace Transform of $x(t) = te^{-2t}u(t)$ . State the ROC and verify the result using Laplace Transform properties.                                                | 5     |  |

|     |                                                                                                                                                                                                                                         |   |     |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|
| 4c  | <p>Explain unilateral Laplace Transform. Using Laplace Transform, obtain the transfer function and impulse response of an LTI system</p> $\frac{dy(t)}{dt} + 3y(t) = 2x(t)$ <p>and state the stability and causality of the system.</p> | 5 | 204 |
| Q 5 | Attempt any 2 parts of the following                                                                                                                                                                                                    |   |     |
| 5a  | Define the Z-Transform. Explain the Region of Convergence (ROC) and discuss its properties with suitable examples.                                                                                                                      | 5 | 005 |
| 5b  | Explain the relationship among the Laplace Transform, Fourier Transform and Z-Transform. Discuss how the ROC determines the stability and causality of a discrete-time LTI system.                                                      | 5 | 005 |
| 5c  | <p>Find the inverse Z-transform of</p> $X(z) = \frac{z}{(z-1)(z-2)}$ <p>using partial fraction expansion.</p>                                                                                                                           | 5 | 005 |